

Classifier-Causal Mask Discovery and Trust-Region Guidance for Localized Diffusion Counterfactuals

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Abstract

Visual counterfactuals should identify where a classifier can be interventionally affected and then change that evidence without unnecessary identity, locality, or semantic drift. Saliency alone provides correlational localization, while permanently weighted diffusion guidance cannot reliably express target-first priorities as denoising geometry changes. We introduce a training-free framework that verifies semantic regions through paired same-seed diffusion interventions, freezes their effects in a classifier-specific influence graph, and resolves a minimal semantic support for each source. Within that support, adaptive lexicographic trust-region causal concept intervention (ADAPTIVE TR-CCI) measures the predicted-clean image, requests attainable target progress, minimizes the best achievable identity/locality violation, and only then minimizes non-target drift. On the full paired end-to-end evaluation cohort, ADAPTIVE TR-CCI reaches 81.0% FR, COUT 0.1152, and CD 2.8894, while retaining 100.0% FVA, FS 0.9958, FID 17.4345, and symmetric FID 72.3917. Relative to BLD under the same semantic mask and attack protocol, it gains 16.7 percentage points in FR and 0.2134 in COUT and reduces CD by 1.8%. The causal claim is explicitly limited to the frozen classifier-generator system rather than the physical world.

Keywords: visual counterfactuals, causal intervention, semantic mask discovery, diffusion guidance, trust-region optimization

1 Introduction

A visual counterfactual explains a classifier by showing how an input could be changed to obtain a desired output. For faces, crossing the target boundary is not enough: the result should preserve the person’s identity, avoid changing unrelated predicted attributes, and restrict visible modifications to relevant facial evidence. Otherwise, a plausible image may still be an unfaithful explanation of the model.

Localized diffusion counterfactuals face two dependent problems. First, localization must determine *where* the generator is permitted to act. Gradient saliency can identify pixels associated with a decision, but association does not show that modifying a highlighted semantic component changes that decision. Second, optimization must determine *how* to edit the chosen region. A weighted

objective

$$L = w_T L_T + w_D L_D + w_I L_I + w_L L_L \quad (1)$$

must use one set of weights across changing denoising geometry and incompatible states: reaching the target and preserving it after success. Strong preservation may block the target, whereas strong target guidance can move unrelated classifier outputs.

We connect these decisions in one training-free framework. Controlled masked diffusion interventions verify which semantic regions influence the explained classifier and form a frozen counterfactual influence graph. At inference, the graph constrains source evidence to a minimal verified semantic support. That support defines where change is allowed; lexicographic trust-region CCI then determines how to change it. Target progress is primary, identity and outside-mask locality form the best attainable safety envelope, and non-target drift is reduced only inside that envelope.

Figure 1 visualizes the complete method. Offline controlled interventions establish which semantic regions can affect the classifier; online source evidence resolves that policy into a mask before adaptive CCI generates the localized counterfactual. The image panels are directly exported from the paired evaluation, not synthesized or retouched for presentation.

Our contributions are:

- classifier-causal semantic mask discovery through paired same-seed region interventions and confidence-based influence-graph verification;
- source-specific minimal region resolution using a frozen graph and source evidence, without inspecting or reranking generated candidates;
- predicted-clean lexicographic trust-region guidance and preservation-aware final restoration confined to the resolved support, improving FR by 16.7 points and COUT by 0.2134 in paired end-to-end evaluation.

Here, “causal” denotes an interventional effect inside a fixed explained classifier and generator. It does not identify biological, anatomical, social, or other real-world causes.

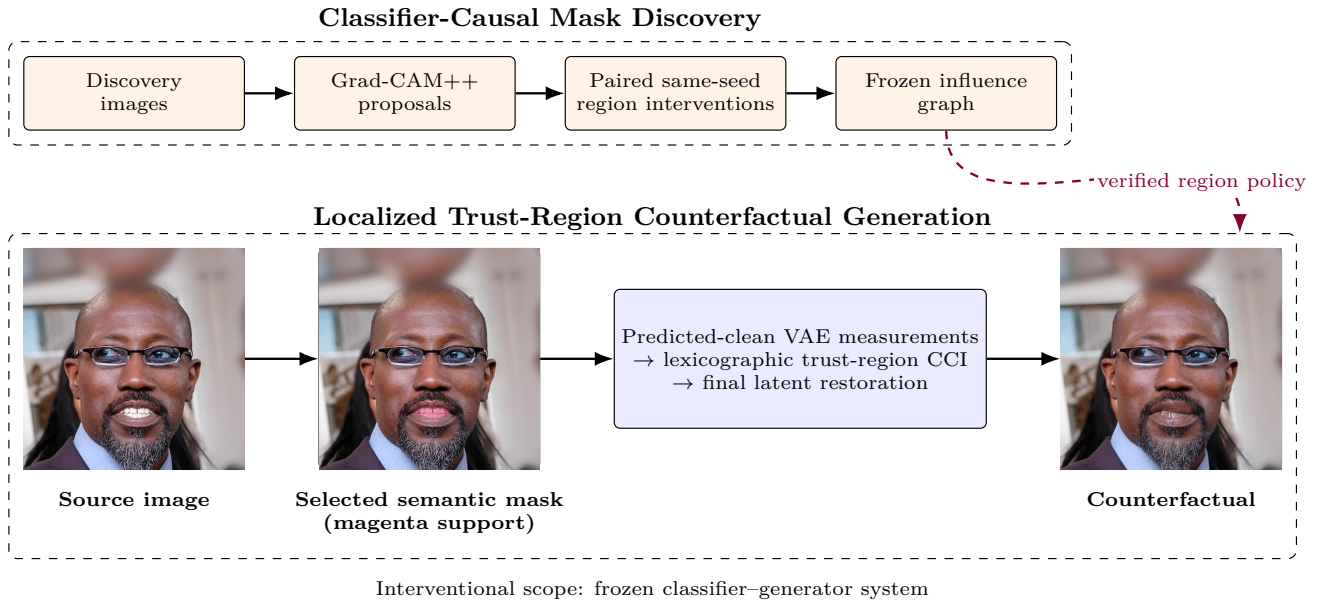


Figure 1: End-to-end method overview. The offline discovery stage turns saliency proposals into a verified classifier-specific influence graph. For a new source, the frozen policy determines the semantic intervention mask; predicted-clean lexicographic guidance and final latent restoration then produce one localized counterfactual. The real image panels are unretouched evaluation artifacts apart from resizing and the magenta mask overlay. Causal claims are limited to the frozen classifier-generator system.

2 Related Work

Diffusion counterfactual explanations. Generative counterfactual methods use structured latent spaces to produce plausible alternatives [15, 8]. DiME and ACE combine classifier objectives with diffusion image priors [9, 10]; TIME and ECED extend diffusion counterfactual construction [11, 14]; and DiG-IN optimizes diffusion variables for model-analysis objectives [1]. A prior facial-counterfactual system combines face-part masks, diffusion generation, and adversarial analysis [20]. Our focus is the connection between intervention-verified localization and priority-constrained generation.

Saliency and mask-based localization. Grad-CAM and Grad-CAM++ localize classifier evidence through gradients [18, 5], but their maps remain observational explanations. Blended latent diffusion (BLD) confines synthesis by restoring a correspondingly noised source latent outside a spatial mask after each scheduler step [2]. MaskDiME derives dynamic classifier-gradient masks during denoising [7]. In contrast, we use Grad-CAM++ only to screen and resolve semantic candidates; causal verification comes from paired diffusion interventions, and the selected support is frozen before counterfactual generation.

Predicted-clean and multi-objective guidance. Latent diffusion samples in an autoencoder space [16], and DDIM supplies the scheduler used here [19]. Universal Guidance and diffusion posterior sampling demon-

strate external measurements on predicted-clean estimates [3, 6]. Gradient-surgery methods modify conflicting multi-task directions [21], but a weighted or projected direction still leaves the priority policy implicit. We express that policy as sequential local subproblems: attain target progress, minimize unavoidable safety violation, then reduce collateral classifier drift.

3 Methodology

3.1 Preliminaries

Let $x_i \in [0, 1]^{3 \times H \times W}$ be a source image, $f_j(x)$ the logit of attribute j , T the explained target, and $y_T \in \{0, 1\}$ its desired value. The desired probability is $p^*(x) = \sigma((2y_T - 1)f_T(x))$. Let $\mathcal{R}$ denote semantic component regions and $R \subseteq \mathcal{R}$ an allowed region set.

For fixed generation seed s , define the paired intervention effect

$$\Delta_{i,s,R} = p^*\left(x_{i,s}^{\text{do}(R)}\right) - p^*(x_i). \quad (2)$$

The operation $\text{do}(R)$ permits the generator to modify exactly R while source, prompt, seed, model, scheduler, and generation settings are held fixed. This is a classifier-generator intervention, not a claim that the semantic region causes an attribute in people or in the physical world.

3.2 Overall Framework

Figure 1 summarizes the complete method. A discovery cohort is processed once to screen semantic candi-

dates, estimate controlled intervention effects, and freeze a reusable influence graph. For each new source, the graph and source Grad-CAM++ evidence select one minimal verified semantic region set. Selection occurs before synthesis and yields both a hard semantic mask and a soft generation mask. The former defines the allowed explanatory support; the latter provides smooth latent support for gradients and BLD blending. Predicted-clean lexicographic guidance, DDIM sampling, source blending, and final latent restoration then generate one audited counterfactual without expanding the chosen region.

3.3 Classifier-Causal Mask Discovery

Candidate screening. For each discovery source, Grad-CAM++ is intersected with semantic component masks. Captured attribution mass and within-component density rank plausible regions. This stage reduces the intervention search space but supplies no causal verification: high attribution is treated only as a proposal.

Controlled region interventions. We generate singleton and selected joint region interventions under the paired conditions of Eq. (2). The mask is the only allowed source of spatial variation; prompt, seed, model, scheduler, and inference settings remain identical. Post-generation attack is disabled so the measured effect belongs to localized generation rather than a subsequent optimizer. Alongside desired-probability change, the discovery record stores semantic mask cost, strict outside change, identity, non-target drift, and joint-region synergy.

Influence-graph verification. We aggregate repeated observations with a source-ID-clustered bootstrap, so multiple seeds of the same source are not treated as independent images. A target-to-region edge is verified only when the singleton region has positive mean effect, the lower confidence bound exceeds zero, and the frozen minimum sample count is met. Joint interventions estimate whether a union supplies additional effect beyond its singleton members. The verified edges, costs, confidence intervals, and synergies form a frozen classifier-specific counterfactual influence graph.

Source-specific resolution. For a held-out source, we intersect its Grad-CAM++ map with its semantic masks and consider only regions admitted by the frozen graph. Among verified unions, we choose the smallest-cost set reaching the predeclared saliency-coverage requirement. Resolution can use only the source image, frozen graph, source Grad-CAM++ map, and semantic masks. It cannot inspect generated outputs, rerank candidates, retry a failed explanation, or expand the region after observing the counterfactual. This separation prevents output-dependent mask selection from masquerading as discovery.

3.4 Localized Trust-Region CCI

Localized objective. For required desired probability p_{req} , define

$$\kappa = \log \frac{p_{\text{req}}}{1 - p_{\text{req}}}, \quad c_T(z) = \kappa - (2y_T - 1)f_T(x(z)). \quad (3)$$

The target requirement is met when $c_T(z) \leq 0$. The resolved binary semantic mask m_s states where change is allowed to count, while a dilated, feathered generation mask $m_g \in [0, 1]$ supplies smooth latent support and source blending. Non-target preservation includes all other classifier outputs:

$$D_{\text{NT}}(z) = \frac{1}{K-1} \sum_{j \neq T} \text{Huber}_\delta(p_j(x(z)) - p_j(x_i)). \quad (4)$$

Identity and locality define safety residuals

$$c_I(z) = 1 - \cos(e(x(z)), e(x_i)) - \varepsilon_I, \quad (5)$$

$$c_L(z) = \text{mean}((1 - m_g)|x(z) - x_i|) - \varepsilon_L. \quad (6)$$

Identity uses a frozen FaceNet representation trained on VGGFace2 [17, 4].

Predicted-clean measurements. At denoising step t , let z_t be the current latent, $\epsilon_\theta(z_t, t)$ the detached U-Net prediction, and $\bar{\alpha}_t$ the scheduler cumulative alpha. For epsilon prediction,

$$\hat{z}_0 = \frac{z_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(z_t, t)}{\sqrt{\bar{\alpha}_t}}, \quad (7)$$

$$\hat{x}_0 = D(\hat{z}_0/s_{\text{VAE}}), \quad s_{\text{VAE}} = 0.18215. \quad (8)$$

The target, drift, identity, and locality models evaluate $\hat{x}_0$. Each image-space gradient passes through the fixed VAE decoder,

$$g = J_D(\hat{z}_0)^\top \nabla_x L, \quad (9)$$

but not through the U-Net. The decoder is therefore a differentiable measurement map, not another diffusion transition.

Lexicographic trust-region update. Let $g_T = \nabla c_T$, $g_D = \nabla D_{\text{NT}}$, $g_I = \nabla c_I$, and $g_L = \nabla c_L$ with respect to $\hat{z}_0$. Generation support is applied before norms, inner products, or optimization. If the target is unmet, requested local progress is

$$q_t = \min(\max(c_T, 0), \chi \Delta_t \|g_T\|_2), \quad (10)$$

where Δ_t is the clean-latent trust radius and $0 < \chi < 1$. The first subproblem finds the smallest unavoidable worst safety residual:

$$\tau^* = \min_{d, \tau} \tau \quad (11)$$

$$\begin{aligned} \text{s.t. } & g_T^\top d \leq -q_t, \\ & c_I + g_I^\top d \leq \tau, \quad c_L + g_L^\top d \leq \tau, \\ & \|d\|_2 \leq \Delta_t. \end{aligned} \quad (12)$$

Inside that best envelope, the second subproblem minimizes collateral drift:

$$d^* = \arg \min_d g_D^\top d + \frac{1}{2\eta_t} \|d\|_2^2 \quad (13)$$

$$\begin{aligned} \text{s.t. } & g_T^\top d \leq -q_t, \\ & c_I + g_I^\top d \leq \tau^* + \zeta, \\ & c_L + g_L^\top d \leq \tau^* + \zeta, \\ & \|d\|_2 \leq \Delta_t. \end{aligned} \quad (14)$$

Once the target is met, its progress constraint becomes the guard $c_T + g_T^\top d \leq 0$. Deterministic bisection reduces an infeasible request; an unreliable target gradient skips the update. Because only four masked gradients participate, the subproblems are solved by active-set enumeration in their span using a small float64 Gram matrix. The radius adapts from achieved target progress: it shrinks below one-quarter of the accepted request and may grow when more than three-quarters is realized at the boundary.

The accepted clean displacement maps exactly to epsilon-prediction space:

$$\Delta\epsilon_t = -\sqrt{\frac{\bar{\alpha}_t}{1 - \bar{\alpha}_t}} d^*. \quad (15)$$

After the scheduler step, BLD restores source content outside m_g :

$$z_{t-1} \leftarrow m_g \odot z_{t-1} + (1 - m_g) \odot z_{t-1}^{\text{src}}. \quad (16)$$

A permanent weighted composition applies the same nominal gradient mixture at every iterate. In contrast, ADAPTIVE TR-CCI solves Eqs. (12)–(14) at the current iterate, so its effective coefficients follow the current residuals and gradient geometry.

Preservation-aware final restoration. The final post-BLD latent may remain below the target threshold because a local step is attenuated by the scheduler or source blend. We therefore run at most 12 restoration iterations on that actual latent. Each iteration recomputes the two lexicographic subproblems through the fixed decoder and evaluators and tests fractions $\{1, 1/2, 1/4, 1/8\}$ of the proposed step. Before target success, acceptance requires strictly improved desired probability without worsening the best identity/locality envelope beyond tolerance. After target and safety success, it must maintain both and strictly decrease non-target drift. Every accepted step remains inside a cumulative clean-latent radius; failure of every fraction stops restoration. This stage uses no new U-Net prediction or scheduler transition. The saved uint8 image is evaluated again because VAE decoding and quantization can move a boundary-adjacent result.

4 Experimentation

4.1 Experimental Protocol

Data and task. We evaluate $N = 300$ eligible CelebA-HQ images with CelebAMask-HQ semantic annotations [13, 12] for smile removal (**Smiling** $1 \rightarrow 0$). Source IDs are sampled once and paired exactly across methods. The benchmark fixes the semantic mouth component for every pair, isolating counterfactual optimization after semantic-support resolution. Raw BLD and ADAPTIVE TR-CCI receive the same source, mask, prompt, seed 42, diffusion checkpoint, scheduler, and localized smooth-boundary attack. No output-dependent reranking or region expansion is permitted.

Generation and attack. We use Stable Diffusion 2.1 base with DDIM at 512^2 resolution, 35 denoising steps, classifier-free guidance 5.0, blend-start fraction 0.25, and a mouth mask dilated by eight pixels. ADAPTIVE TR-CCI uses target threshold 0.8, clean-latent radii 0.15/0.01/0.30, target-progress fraction 0.9, reliability threshold 0.10, Huber transition 0.02, at most 12 final-restoration iterations, and cumulative restoration radius 1.8. The post-generation attack is confined to the localized support and follows a fixed perturbation schedule with boundary margin 0.03. The same frozen 40-attribute CelebA classifier guides the attack and CCI and is intentionally the source of truth for all classifier-dependent metrics.

Metrics. FID measures source/output distribution distance using Inception features. Symmetric FID (sFID) averages two deterministic cross-split Fréchet distances to reduce split-direction dependence. FVA is the fraction of paired VGGFace2 embeddings with cosine similarity above 0.5, while FS is paired SimSiam cosine similarity. FR is directional target success at the classifier’s 0.5 decision boundary. MNAC counts non-target binary attribute flips. CD sums absolute correlations between each attribute change and the target-change vector. COUT integrates desired-versus-source evidence while output pixels are inserted in descending change magnitude. Thus FR, MNAC, CD, and COUT consistently explain the same frozen multi-label classifier used by the method; they are not claims of agreement with an independent human or physical-world oracle.

4.2 Quantitative Results

Table 1 shows the central result. ADAPTIVE TR-CCI raises FR from 64.3% to 81.0%, a gain of 16.7 percentage points, and moves COUT from -0.0982 to 0.1152 (+0.2134). It also reduces CD from 2.9437 to 2.8894, a 1.8% reduction, while symmetric FID improves slightly from 72.5453 to 72.3917.

Distributional and identity preservation remain closely matched: FID is 17.3720 versus 17.4345, both methods

Table 1: Paired end-to-end smile-removal results after localized attack. sFID denotes deterministic symmetric FID and FR is in percent. Bold marks the better value; identical FVA values are both bold.

Metric	BLD	ADAPTIVE TR-CCI (Ours)
FID ↓	17.3720	17.4345
sFID ↓	72.5453	72.3917
FVA ↑	100.0	100.0
FS ↑	0.9961	0.9958
MNAC ↓	2.4967	2.6233
CD ↓	2.9437	2.8894
COUT ↑	-0.0982	0.1152
FR (%) ↑	64.3	81.0

reach 100.0% FVA, and FS differs by less than 4×10^{-4} before rounding. MNAC increases from 2.4967 to 2.6233, showing that the large target-success gain does not improve every collateral metric. Taken together, the measurements support a specific claim: adaptive target-first control substantially improves the desired classifier intervention while preserving perceptual quality and reducing correlation-based collateral change.

Published benchmark context. Table 2 positions our verified result beside the CelebA-HQ Smile block reported by MaskDiME, but the values are not a head-to-head experiment. MaskDiME reports ten-repeat split-FID under its 256^2 CelebA-HQ protocol, whereas our sFID is deterministic two-direction split-FID under a 512^2 attacked CelebA-HQ protocol with CelebAMask-HQ annotations. Resolution, classifier, cohort construction, attack, and generation settings also differ. The table is therefore cross-protocol context; Table 1 remains the controlled comparison that isolates adaptive CCI.

4.3 Qualitative Results

Figure 2 complements the aggregate metrics with direct paired comparisons. Raw BLD already confines the edit spatially, but its generated mouth can retain stronger smile evidence. Adaptive CCI produces a more neutral mouth while leaving face shape, hair, clothing, and background visually stable. Full-face panels are shown alongside, rather than replaced by, mouth crops: locality is meaningful only when the edited region is judged in the context of the complete identity. Because every panel comes from the evaluated cohort and shares the pair’s source and settings, the visualization illustrates the same comparison measured by Table 1.

4.4 Discussion

The result pattern reflects the lexicographic design. FR and COUT directly measure progress toward the desired classifier state and show the largest gains. CD also falls,

suggesting that adaptive updates better decouple the target transition from correlated attribute changes. By contrast, FID, FVA, and FS are nearly unchanged because both methods inherit the same diffusion prior and spatial blend. MNAC counts hard threshold crossings and is slightly higher even as CD decreases; the two collateral metrics therefore capture different behavior and should not be conflated.

The causal interpretation remains deliberately narrow. Paired interventions identify effects under a frozen classifier, generator, prompt, and scheduler; they do not establish that a facial component causes an expression in people. Likewise, using one classifier for guidance and classifier-dependent evaluation measures faithfulness to the explained model, not transfer to another model. The local linearizations provide an auditable priority rule rather than a global convergence guarantee, and repeated VAE/evaluator differentiation costs more than raw BLD. Finally, face images and identity embeddings are biometric data; the intended use is consented model auditing with appropriate data protection.

Table 2: Published CelebA-HQ Smile context reported by MaskDiME [7]. The upper block transcribes its 256² benchmark values; the final row reports our 512² attacked CelebA-HQ protocol with CelebAMask-HQ annotations using generated macros. Because protocols and sFID estimators differ, the separator denotes cross-protocol context rather than a controlled ranking. FVA and FR are percentages.

Method	FID ↓	sFID ↓	FVA ↑	FS ↑	MNAC ↓	CD ↓	COUT ↑	FR (%) ↑
<i>Published MaskDiME CelebA-HQ protocol</i>								
DiVE	107.0	—	35.7	—	7.41	—	—	—
STEEX	21.9	—	97.6	—	5.27	—	—	—
DiME	18.10	27.7	96.7	0.67	2.63	1.82	0.65	97.0
ACE ℓ_1	3.21	20.2	100.0	0.89	1.56	2.61	0.55	95.0
ACE ℓ_2	6.93	22.0	100.0	0.84	1.87	2.21	0.60	95.0
LDCE-txt	13.6	25.8	99.1	0.76	2.44	1.68	0.34	—
TiME	10.98	23.8	96.6	0.79	2.97	2.32	0.63	97.1
RCSB	3.04	20.0	100.0	0.93	1.22	3.22	0.83	98.9
MaskDiME	2.51	18.1	100.0	0.94	1.41	2.67	0.69	99.4
<i>Our attacked CelebA-HQ protocol</i>								
ADAPTIVE TR-CCI (Ours)	17.4345	72.3917	100.0	0.9958	2.6233	2.8894	0.1152	81.0

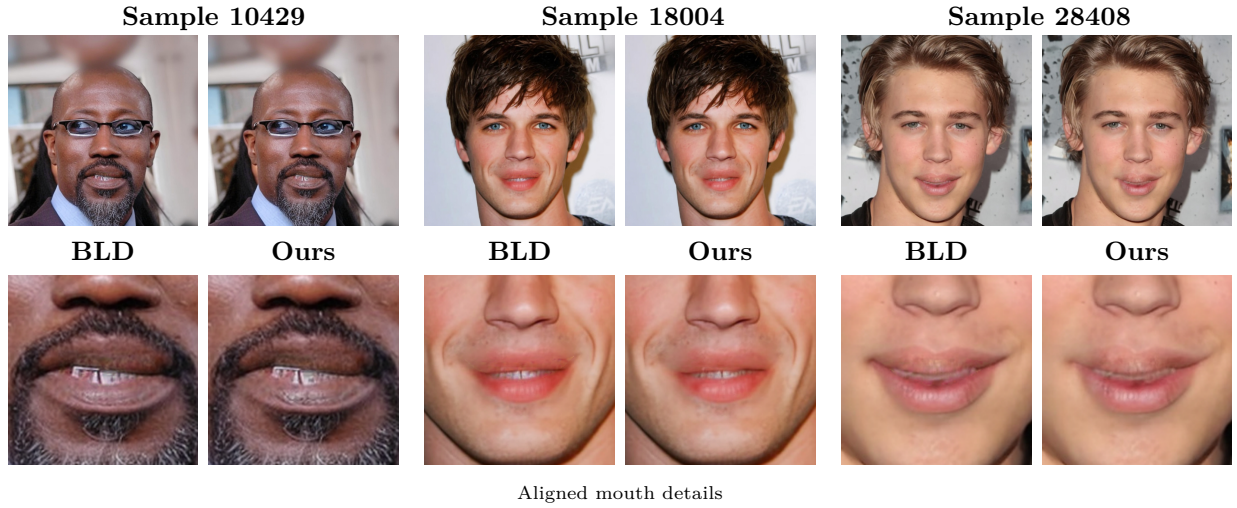


Figure 2: Paired qualitative comparison. Within each pair, raw BLD and adaptive CCI share the source, semantic mask, prompt, seed, scheduler, localized attack, and generation settings. Full faces are shown so the localized mouth edit can be judged together with identity and background preservation; aligned mouth details expose the target evidence at print scale. Images are unretouched evaluation artifacts apart from resizing and cropping.

5 Conclusions

We presented a unified framework in which intervention-verified localization determines where a counterfactual may change and adaptive lexicographic trust-region guidance determines how it changes. Paired same-seed interventions turn saliency-screened semantic components into a reusable classifier-specific influence graph; source evidence then resolves a minimal semantic support without inspecting generated candidates. Within that support, predicted-clean measurements and clean-coordinate trust regions make target, safety, and collateral priorities explicit, including at the final post-blend latent.

In paired end-to-end smile removal, the proposed method improves FR by 16.7 percentage points and COUT by 0.2134, reduces CD by 1.8%, and preserves matched FVA with nearly unchanged FID and FS. These results support localized, model-faithful counterfactual control under attack. They remain causal evidence about

the frozen classifier-generator system, not a statement about physical or human causation.

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